

Volatile metabolite profiling to detect and discriminate stem-end rot and anthracnose diseases of mango fruits

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The volatile metabolites from the headspace gas of containerised mango (*Mangifera indica*) cv. Tommy Atkins fruits, surface wounded and inoculated with the two fungal anamorphic pathogens *Colletotrichum gloeosporioides* and *Lasiodiplodia theobromae*, or non-inoculated (controls), were profiled using a portable gas chromatograph/mass spectrometer to discriminate diseases of mango. Thirty-four compounds were detected relatively consistently among replicates. Several of these were disease/inoculation-discriminatory and were classified into three groups: (i) compounds unique to only one treatment; (ii) compounds common to two or more treatments, but not to all; and (iii) compounds common to all treatments, but varying in their abundance. Two compounds, 1-pentanol and ethyl boronate, were detected in *L. theobromae*-inoculated mangoes alone, while thujol was observed only in *C. gloeosporioides*-inoculated mangoes. Discriminant analysis models based on the abundance of significant mass ions and consistent compounds correctly classified diseases/inoculations in up to 100% of cases. The disease-discriminatory compounds and discriminant analysis models developed here have the potential to be used in the early detection of postharvest diseases of mango fruits after validation under commercial conditions.

Keywords: *Colletotrichum gloeosporioides*, discriminant analysis, disease detection, gas chromatography/mass spectrometry, *Lasiodiplodia theobromae*, postharvest disease

Introduction

Mangoes (*Mangifera indica*) are grown in tropical and subtropical regions around the world (Ploetz *et al.*, 1994). Because of their delicious taste, nutritional value, attractive fragrance and beautiful colour, mangoes are now important fruits in the world market (Mitra, 1997). World mango production in 2002 was estimated at 25 755 000 tonnes, of which two-thirds were produced in India. The biggest exporter of mangoes in 2002 was Mexico (195 000 t), while the USA remained the biggest importer (238 000 t) (Anonymous, 2002).

The world mango trade is expanding, in spite of postharvest losses caused by diseases. Several diseases, such as anthracnose (*Colletotrichum gloeosporioides*), alternaria rot (*Alternaria alternata*), stem-end rot (caused by a number of fungi, including *Lasiodiplodia theobromae*), black mould rot (*Aspergillus niger*) and bacterial black spot or canker (*Xanthomonas campestris* pv. *mangiferaeindicae*), can cause severe postharvest losses during handling and storage. Among these, anthracnose

and stem-end rot are the most prevalent (Barkai-Golan, 2001). Even though infection often occurs in the field and the visibly diseased fruits are culled before storage, latent infections often escape inspection and continue to develop during storage. What is needed is a technology that enables store managers to prevent diseased fruits being shipped long-distance or stored for prolonged periods.

Currently, store managers detect diseases based on visible moulds and disease symptoms, or unpleasant odours produced by rotting fruits. Factors enhancing disease development, such as condensation on the crop or the ceiling of the store, and excessive moisture on the floor or sides of pallet boxes, also have been used as indicators of potential disease progress. However, in postharvest systems significant losses have often occurred by the time the disease is detected by these methods. With advances in biotechnology, more sophisticated products and techniques, such as the enzyme-linked immunosorbent assay (ELISA) (Dewey & Thornton, 1995) and other molecular diagnostic tools, such as the polymerase chain reaction (PCR) and the reverse transcriptase polymerase chain reaction (RT-PCR), complement simple laboratory procedures to diagnose diseases (Schaad *et al.*, 2003). Even though these molecular tools are quite sensitive, their need for destructive collection of a large

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number of samples and the difficulty in collecting from large stores where boxes are piled up high, limit their practicality.

Plants are known to produce hundreds of metabolites (Dixon *et al.*, 2002). Following pathogen attack, host metabolism is altered, resulting in either the production of novel compounds or changes in the levels of existing ones (Kushalappa *et al.*, 2002; Vikram *et al.*, 2004a,b; Hamze-hzarghani *et al.*, 2005). Up to 58 disease-discriminatory metabolites were identified in potato cv. Russet Burbank tubers inoculated with soft and dry rot pathogens (Lui *et al.*, 2005). Host, pathogen and environmental factors influence the type and abundance of these volatiles (Pasanen *et al.*, 1996). Despite these variations, diseases of various fruits and vegetables have been discriminated using multivariate analysis models (Prithiviraj *et al.*, 2004; Vikram *et al.*, 2004a,b; Lui *et al.*, 2005). Factor analysis models based on principal components were developed to discriminate five inoculation-agents/diseases of McIntosh apples and five inoculation-agents/diseases of Cortland and Empire apples (Vikram *et al.*, 2004a,b).

The volatile metabolites produced by mangoes belong to several chemical groups: monoterpenes, sesquiterpenes, esters, aldehydes, ketones, alcohols, organic acids, aliphatic hydrocarbons and aromatics (Shibamoto & Tang, 1990). The majority of total aroma volatile compounds from mango are hydrocarbons, monoterpenes or sesquiterpenes (Winterhalter, 1991). The aroma of mango fruits is affected by various factors including cultivar, production environment, fruit maturity at harvest and storage conditions, including storage time, controlled atmosphere storage and chilling injury. Aroma volatile compounds also affect the overall flavour of the fresh and processed fruits (MacLeod & Snyder, 1985; Lalel *et al.*, 2003; Singh *et al.*, 2004). Although there are several reports of production of volatiles in healthy mangoes, no work has yet been published on the production of volatile metabolites by diseased mangoes. The objectives of this study were to identify disease-discriminatory volatile metabolites released from mango fruits inoculated with two fungal pathogens, and to develop other criteria based on metabolic profiles to discriminate diseases of mango.

Materials and methods

Mango fruits

Mango fruits (cv. Tommy Atkins), produced in Mexico, were provided by a mango-distributing company in Canada (Aliments IMEX Foods Inc.). Mature, but still green, fruits of size 12 (average size 370 g), without any visible blemishes, were picked from the store one day before use in each block, at about weekly intervals.

Pathogen culture

Two fungal pathogens, *C. gloeosporioides* (teleomorph: *Glomerella cingulata*) isolate BT1A-2 and *L. theobromae*

(teleomorph: *Botryosphaeria rhodina*) isolate BT2E-1, were used in this study (obtained from Dr R. C. Ploetz, University of Florida, Tropical Research and Educational Centre, Homestead, FL, USA). Stock cultures were stored at 4°C. *Colletotrichum gloeosporioides* was subcultured on oatmeal agar, incubated at 22°C for 5 days and then exposed to near UV (380–200 nm wavelength) black light for one week to induce sporulation, from which a spore suspension of 10^7 conidia mL⁻¹ of water was prepared. *Lasiodiplodia theobromae* was subcultured on potato dextrose agar (PDA; Difco) and was incubated at 22°C for 5 days and then kept under near UV (380–200 nm wavelength) light for 1 week to induce sporulation. Spores were suspended in water to give 10^4 conidia mL⁻¹.

Inoculation and incubation

Pedicels were removed and the fruits surface-sterilized in 1% sodium hypochlorite solution for 15 min, washed, and dried in a laminar flow cabinet. At the equatorial region, six holes (3 mm diameter, 3 mm deep) were made using a sterile cork borer and each hole was inoculated either with 20 µL of *L. theobromae* or 30 µL of *C. gloeosporioides* spore suspension. Mangoes were sprayed with sterile distilled water and placed in 2-L, 110-mm-diameter wide-mouth glass jars containing 13 mL of sterile deionized water in the bottom to provide a saturated atmosphere and facilitate initial establishment of infection. Stainless steel supports were used to suspend mangoes above the water surface. The bottles with fruits were covered with aluminium foil and kept in an incubator at 20 ± 2°C in the dark. After 48 h the aluminium cover was perforated to increase air exchange.

Volatile metabolite sampling and GC/MS analysis

On the day of sampling, each fruit was transferred to a new sampling jar equipped with a magnetic stirrer at the bottom and covered with a 0.02-mm-thick Teflon film (ENFLO Canada Ltd). The bottles were flushed with pure air. After 2.5 h of incubation at 22°C, the accumulated headspace gas was stirred for 5 min, sampled and analysed using a portable GC/MS (Hapsite model; INFICON) (Prithiviraj *et al.*, 2004). A needle connected to the hand-held probe of the GC/MS was inserted into the bottle to sample headspace gas. The GC/MS was programmed to sample headspace gas for 15 s at a rate of 100 mL min⁻¹. The volatile metabolites were pre-concentrated in a tube trap containing 15 mg carboxen (Carbopack-X; INFICON), thermally desorbed at 225°C and compounds/peaks separated in the GC using a 30-m-long SPB-5 capillary column (Supelco). Nitrogen was used as a carrier gas at a flow rate of 3 mL min⁻¹. The column was heated to 50°C followed by a ramping of 3°C min⁻¹ for 50 min to 200°C, where the temperature was maintained for 2 min. The GC was interfaced to a MS equipped with a quadrupole analyser and the mass spectrum was scanned at the rate of 0.9 spectra s⁻¹ over a mass range of 46–300 m/z.

Disease severity assessment

The diameter and the depth of the diseased tissues were measured 6 days after inoculation (DAI) for *C. gloeosporioides* and 4 DAI for *L. theobromae*, from which the volume of diseased tissue was calculated.

Experimental design and data analysis

The experiment was designed as a factorial with four main factors of inoculation: non-wounded and non-inoculated control (N-control), wounded and non-inoculated control (W-control), and inoculated with *C. gloeosporioides* or *L. theobromae*. There were four subfactors of incubation time (2, 3, 4, and 6 DAI) for *C. gloeosporioides* and three subfactors of incubation time (2, 3, and 4 DAI) for *L. theobromae* (to represent different disease severity levels). Each experimental unit consisted of one mango fruit. The entire experimental block was replicated eight times, at weekly intervals. The GC/MS output consisted of scans and the abundance of mass ions (in the range 46–300 *m/z*), which were used to calculate metabolic fingerprints based on the normalized abundance of mass ions or normalized abundance of the relatively consistent metabolites (Vikram *et al.*, 2005).

Metabolite fingerprints and compound identification

The abundance of each mass ion for all scans in a total ion chromatogram were summed and normalized by dividing the abundance of each mass ion by the total for all the 150 mass ions (from 46 to 195 *m/z*; as spectra of most metabolites were within this range) and was designated as a metabolic fingerprint (Vikram *et al.*, 2005). No laborious process of compound identification was necessary to obtain these fingerprints, and thus the technique has the potential for automation. The abundance of 150 mass ions served as 150 specific sensors, similar to those in electronic nose instruments.

The output from GC/MS was exported to an EXCEL spreadsheet where the PIVOT TABLE procedure was used to sort data according to retention time. The spectrum of each peak at a given retention time (± 5 s) was compared manually with the first 15 choices of the NIST library (2.0; INFICON) to select the most suitable spectral match, especially for those ions with high relative abundance. Compounds which occurred in at least four out of 24 observations (eight blocks $\times$ three incubation times) for *L. theobromae* and out of 32 observations (eight blocks $\times$ four incubation times) for *C. gloeosporioides*, N-control and W-control, and with an abundance $>10^3$, were used to calculate the frequency and average abundance of relatively consistent metabolites. A compound that was either unique to a treatment, or common to two or more but not to all treatments was designated as a disease/inoculation-discriminatory metabolite. The abundance of relatively consistent metabolites, excluding those unique to a treatment, were normalized by dividing each by the total abundance for

all the relatively consistent compounds, and designated as the metabolic fingerprint.

Statistical analysis

The metabolite fingerprints based on the normalized abundance of 150 mass ions (range 46–195 *m/z*) and the normalized abundance of 31 relatively consistent compounds were subjected to univariate analyses to select the significant ($P = 0.1$) mass ions and metabolites, respectively. The significant mass ions and metabolites were subjected to step-wise discriminant analysis using the STEPDISC procedure of SAS (Khattree & Naik, 2000); the selected variables were in turn subjected to discriminant analysis using the DISCRIM procedure. Several models were developed for each of the two metabolic fingerprint datasets based on the normalized abundance of mass ions or compounds, which were re-classified into four disease severity groups: (i) low disease severity (2 DAI for *L. theobromae* and 3 DAI for *C. gloeosporioides*, N-control and W-control); (ii) medium disease severity (3 DAI for *L. theobromae* and 4 DAI for *C. gloeosporioides*, N-control and W-control); (iii) high disease severity (4 DAI for *L. theobromae* and 6 DAI for *C. gloeosporioides*, N-control and W-control); and (iv) low + medium + high disease severity (all disease severity levels).

The objective of the discriminant analysis was to find mathematical rules to determine to which group an observation belonged (Johnson, 1998). Posterior probability of an observation for each of the competing inoculations/diseases was used to classify the observation into the disease/inoculation that gave the largest posterior probability. A perfect classification had a posterior probability equal to prior probability of 25% for each of the four inoculations/diseases. The probabilities of misclassification estimates were used to test discriminant rules to correctly classify the observations into inoculations/diseases. To estimate misclassification probability, a cross-validation procedure, or jackknifing, was used. The sequence of the procedure started with removing the first observation vector; based on the remaining data a new discriminant rule was formed. This new rule was then used to classify the first removed observation to test if it was correctly classified. A summary table of cross-validated data was created by repeating the same cycle with the rest of the observation vectors. In addition to cross-validation, test-validation was conducted using the DISCRIM TESTDATA procedure of SAS and a dataset that was not used in developing the model to estimate the probability of identifying unknown fingerprints.

Results

Disease progress

Disease severity varied among pathogen inoculations and incubation times. No visible disease symptoms were observed 2 DAI in fruits inoculated with either of the pathogens. Visible disease symptoms were observed 3 DAI, increasing with increase in incubation time. The lesions caused by *L.*

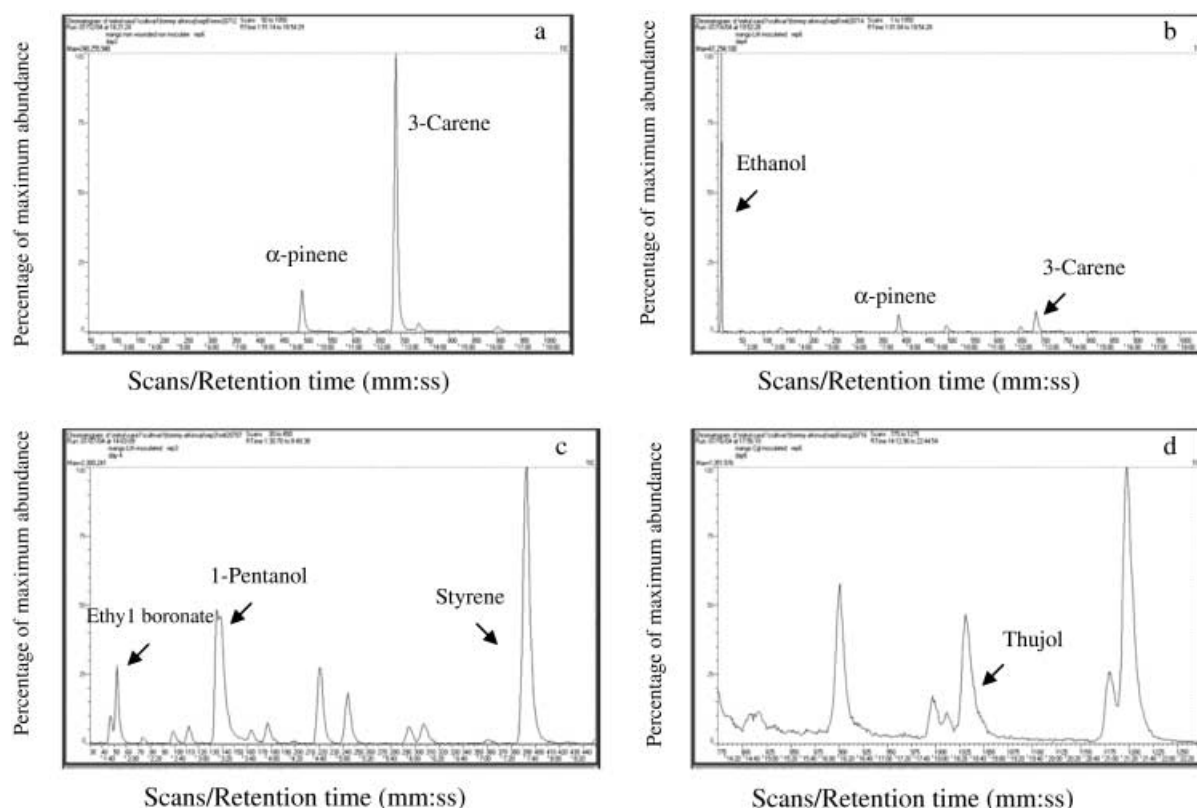


Figure 1 Sections of total ion chromatograms showing specific peaks with high abundance (percentage of maximum abundance at scans or retention times – minutes (mm) and seconds (ss)) of unique compounds. (a) N-control (non-inoculated), 4 days after inoculation (DAI), with peaks of the two most abundant compounds: 3-carene and α -pinene; (b) *Lasiodiplodia theobromae*-inoculated mangoes, 4 DAI, showing higher abundance of ethanol compared with other compounds such as 3-carene and α -pinene; (c) *L. theobromae*-inoculated mangoes, 4 DAI, showing two unique compounds: 1-pentanol and ethyl boronate; (d) *Colletotrichum gloeosporioides*-inoculated mangoes, 6 DAI, showing the presence of one unique compound, thujol.

theobromae and *C. gloeosporioides* were 2.5 and 1 cm in diameter, respectively, at 3 DAI, increasing to 5 and 1.2 cm, respectively, at 4 DAI, whilst the diameter of *C. gloeosporioides* lesions increased to 2.2 cm at 6 DAI. The non-wounded control remained without any disease throughout the 6-day incubation period, while the wounded control showed slight browning at the inoculation site, but no spread of this symptom was noticed. The average volume of diseased tissue was 19.6 cm³ for *L. theobromae* at 4 DAI, and 4.5 cm³ for *C. gloeosporioides* at 6 DAI.

Volatile metabolic profiles

The number of volatile compounds detected in the headspace of the mango fruits varied among inoculations/diseases, incubation times and experimental blocks. In total, more than 200 peaks were detected. Only 34 volatile metabolites (Table 1), high in abundance and relatively consistent over the replicates, were tentatively identified using the NIST library (compound mass spectra available at McGill-MD, <http://www.metabolomics.mcgill.ca>). The relative abundance of metabolites varied from 5.6×10^3 to 5×10^8 . Most of the volatiles eluted within 20 min. The

number of compounds detected was 76 at 2 DAI, increasing to 103 at 4 DAI, then decreasing to 73 at 6 DAI. The highest number of volatile compounds was detected in the N-control. The W-control, *L. theobromae* and *C. gloeosporioides* treatments mangoes yielded 81, 77 and 60 compounds, respectively.

The relatively consistent metabolites that occurred most frequently in all treatments were: 3-carene, α -pinene, limonene, camphene, β -pinene, α -phellandrene, toluene, ethyl benzene, ethyl acetate, borane methyl sulfide complex, methyl 2-methyl-2-propenoate, chloroform, 1-methyl-4-(1-methyl ethylidene) cyclohexene, 2,6-dimethyl-2,4,6-octatriene, methylcyclohexane, 3-pentanone, 1,3-dimethyl-cis-cyclohexane, 2-methylfuran, (E)-3-carene-2-ol; methylene chloride and ethyl hexanoate (Table 1).

In the majority of cases, the compound 3-carene had the highest abundance in all of the treatments; the maximum abundance was found in the W-control with 1.3×10^9 . α -Pinene was the second most abundant compound in all the treatments; its maximum abundance was observed in the W-control (3.4×10^8 ; Fig. 1a). In some cases when disease severity was high, ethanol was detected in the greatest abundance, indicating the generation of ethanol in well-progressed disease (Fig. 1b).

Table 1 Volatile metabolites, their average (across replicates and incubation times) relative abundance ($\times 10^5$) and frequency of occurrence (in parenthesis; out of 32 for all except for *L. theobromae* = 24), detected in the headspace gas of mango fruits (cv. Tommy Atkins) non-inoculated (N-control), wounded and inoculated with water (W = W-control) or *Colletotrichum gloeosporioides* (C) or *L. theobromae* (L.)

Compounds	<i>C. gloeosporioides</i>	<i>L. theobromae</i>	N-control	W-control	Agent ^a
1-Pentanol		43.38 (16)			L
Ethyl boronate		9.38 (11)			L
Thujol	6.70 (7)				C
1-Butanol	1.73 (6)	2.78 (5)			CL
Ethyl propanoate	2.21 (7)	5.00 (15)			CL
Styrene	11.69 (26)	129.11 (16)			CL
Methyl octanoate	0.12 (4)		0.74 (3)	0.53 (3)	CNW
2,6-dimethyl-2,4,6-octatriene	31.80 (30)	40.49 (12)	11.29 (22)	44.45 (29)	CLNW
Ethyl 2-butenate	5.31 (9)	10.93 (11)	0.06 (2)	0.59 (3)	CLNW
Methyl (Z)-2-butenate	4.83 (15)	0.02 (1)	0.59 (7)	1.51 (8)	CLNW
Methyl 2-methyl-2-propenoate	79.21 (18)	63.86 (13)	17.88 (6)	17.43 (9)	CLNW
3-Carene	8215.18 (32)	8019.6 (24)	4596.4 (31)	13 456.4 (31)	CLNW
3-Pentanone	0.23 (14)	0.63 (10)	0.35 (10)	0.32 (18)	CLNW
α -Phellandrene	15.59 (11)	18.44 (9)	27.18 (24)	47.23 (21)	CLNW
α -Pinene	1894.34 (32)	2619.90 (24)	1148.36 (32)	3428.46 (32)	CLNW
β -Pinene	145.25 (32)	174.31 (23)	95.65 (31)	232.95 (31)	CLNW
Borane-methyl sulfide-complex	16.75 (20)	1.25 (7)	12.74 (19)	15.62 (19)	CLNW
Ethyl 2-methyl-butanoate	0.40 (5)	5.96 (10)	0.04 (1)	0.02 (4)	CLNW
Ethyl butanoate	39.71 (11)	62.25 (13)	2.36 (4)	3.78 (5)	CLNW
Camphene	46.58 (23)	73.78 (15)	40.12 (32)	91.49 (26)	CLNW
Chloroform	0.73 (4)	1.74 (8)	1.07 (9)	1.03 (6)	CLNW
1,3-Dimethyl- <i>cis</i> -cyclohexane	1.39 (3)	2.25 (10)	3.07 (19)	1.14 (11)	CLNW
Methylcyclohexane	3.18 (9)	2.95 (6)	3.59 (13)	2.27 (9)	CLNW
1-Methyl-4-(1-methylethylidene) cyclohexene	189.98(32)	198.97 (24)	92.77 (32)	362.99 (31)	CLNW
(E)-3-Carene-2-ol	79.42 (24)	36.28 (12)	44.64 (32)	75.51 (21)	CLNW
Ethanol	310.68 (32)	1784.18 (24)	95.23 (19)	14.07 (27)	CLNW
Ethyl acetate	8.87 (7)	11.76 (9)	0.20 (1)	14.13 (3)	CLNW
Ethylbenzene	0.06 (2)	0.14 (3)	0.13 (3)	0.11 (3)	CLNW
2-Methylfuran	4.77	0.7 (4)	1.6 (15)	2.5 (17)	CLNW
Ethyl hexanoate	131.7 (12)	100.7 (11)	13.2 (8)	42.5 (7)	CLNW
Limonene	392.6 (32)	398.8 (24)	192.4 (31)	646.6 (31)	CLNW
Methylene chloride	0.2 (3)	0.2 (4)	0.1 (2)	0.3 (4)	CLNW
Ethyl octanoate	14.2 (6)	4.8 (9)	0.4 (3)	1.9 (5)	CLNW
Toluene	19.1 (32)	11.2 (19)	11.1 (31)	9.9 (30)	CLNW

^aC, inoculated with *C. gloeosporioides*; L, inoculated with *L. theobromae*; N, non-inoculated control; W, water-inoculated control.

Disease/inoculation-specific volatile metabolites

Among the 34 relatively consistent metabolites detected, 13 were disease-discriminatory. Depending on the specificity of these compounds three classes were identified: (i) unique compounds, i.e. metabolites detected in only one treatment; (ii) compounds common to two or three treatments, but not to all; and (iii) compounds common to all treatments, but with significant variation in their abundance.

Disease discrimination based on unique compounds

1-Pentanol and ethyl boronate were unique to *L. theobromae* (Fig. 1c; Table 1) while thujol was unique to *C. gloeosporioides* (Fig. 1d; Table 1). The abundance and frequencies of occurrence of these compounds were highest at 3 and 4 DAI in *L. theobromae* and at 6 DAI in *C. gloeosporioides*, than at other DAIs.

Disease discrimination based on compounds common to a few but not all treatments

1-Butanol, ethyl propanoate and styrene were detected only in pathogen-inoculated mangoes, not in N-controls

and W-controls, and thus could be used to discriminate diseased from healthy fruits. Ethyl octanoate was detected from *C. gloeosporioides*, N-control and W-control treatments, but not from *L. theobromae*-inoculated mangoes, indicating the absence of stem-end rot (Table 1).

Diseases discrimination based on abundance of compound

Ethanol was detected in all the treatments, but its abundance was highest in *L. theobromae*-inoculated and moderately high in *C. gloeosporioides*-inoculated fruits, but very low in controls (Fig. 2a). Abundance of ethyl butanoate was significantly higher in *L. theobromae* and *C. gloeosporioides* treatments than in controls (Table 1; Fig. 2b). The onset of the appearance of this compound was earlier in *L. theobromae*-inoculated fruits (2 DAI) than in *C. gloeosporioides*-inoculated fruits (4 DAI). In *L. theobromae*-inoculated fruits, the abundance of ethyl 2-methyl-butanoate reached 1.7×10^6 at 4 DAI, compared with only 0.2×10^6 at 6 DAI in *C. gloeosporioides*-inoculated fruits, while it remained low in controls (Fig. 2c). The abundance of ethyl octanoate at 4 DAI

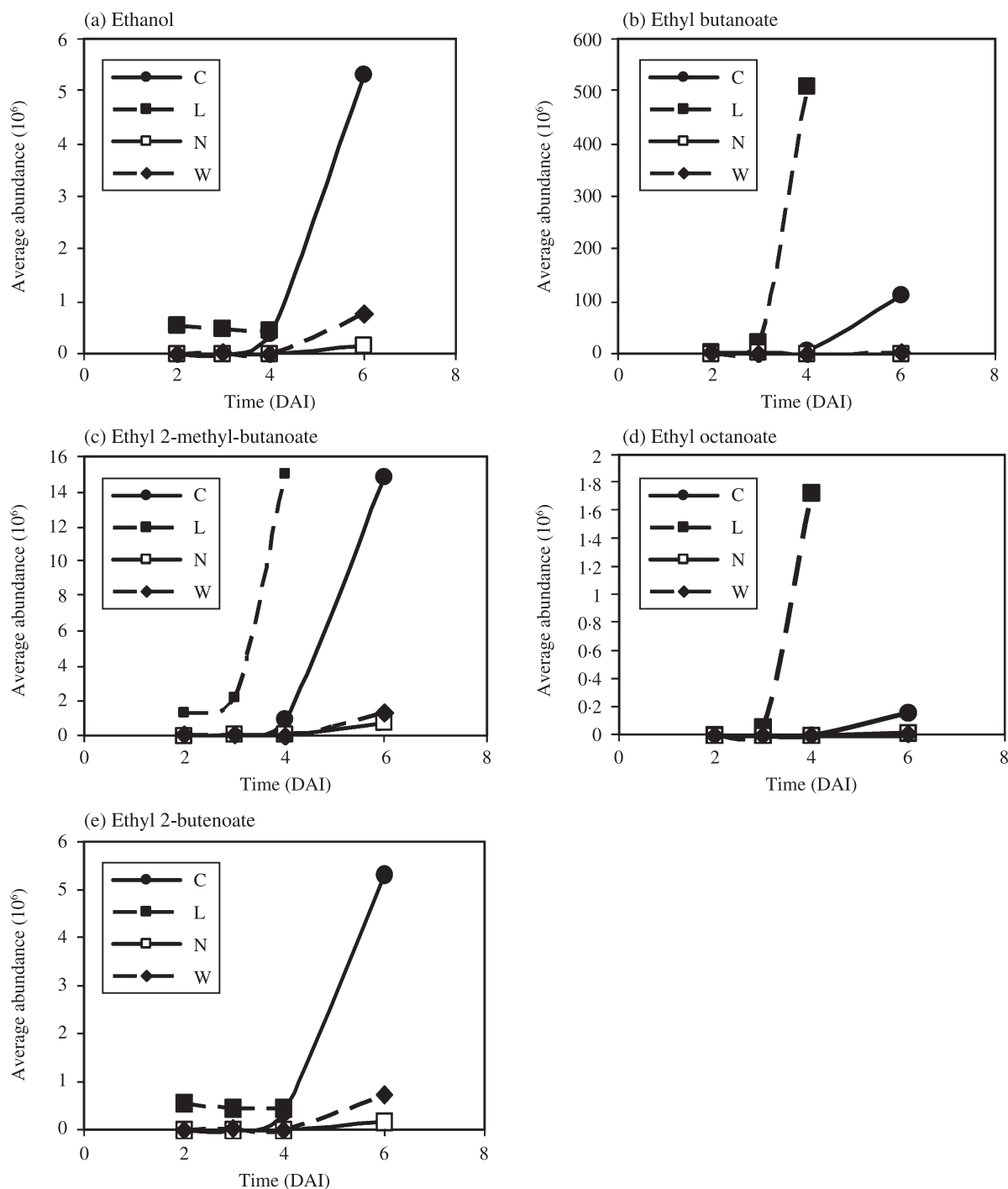


Figure 2 Average abundance of disease-discriminatory compounds at different incubation periods (DAIs), following non-inoculation (N) or inoculation of mangoes with water (W), *L. theobromae theobromae* (L) or *C. gloeosporioides* (C).

was relatively high in *L. theobromae*-inoculated fruits compared to *C. gloeosporioides*-inoculated fruits, where it stayed almost constant at 0.5×10^6 over the incubation time, until 4 DAI (Fig. 2d). The abundance of ethyl octanoate was low in controls. In the case of ethyl 2-butenate it was highest in the *C. gloeosporioides* treatment (1.8×10^6 at 6 DAI; Fig. 2e).

Models to discriminate diseases/inoculations based on abundance of mass ions

Metabolic fingerprints based on the abundance of mass ions (46–195 *m/z*) were grouped according to disease severity into (i) low, (ii) medium, (iii) high and (iv) all disease severity groups for discriminant analysis model development.

Table 2 Percentage of volatile metabolic fingerprints correctly classified into inoculation treatments by re-substitution (DA models), and cross-validation of DA models by discriminant analysis of metabolic fingerprints based on normalized abundance of significant mass ions (selected from 150 mass ions in the range 46–195 *m/z*) or of significant metabolites (selected from 31 relatively consistent metabolites) of mango fruits non-inoculated (N-control), wounded and inoculated with water (W-control), *Colletotrichum gloeosporioides* or *Lasiodiplodia theobromae*. Models were developed separately for low, medium, high and all (low, medium and high) disease severity levels

Disease severity	DA model/validation	<i>C. gloeosporioides</i>	<i>L. theobromae</i>	N-control	W-control	Error
Mass ions						
Low	Re-substitution	13	63	25	88	0.53
	Cross-validation	13	63	25	88	0.53
Medium	Re-substitution	57	38	63	25	0.54
	Cross-validation	57	38	63	13	0.59
High	Re-substitution	100	100	100	100	0
	Cross-validation	100	100	100	88	0.03
All disease severity levels	Re-substitution	71	58	50	71	0.38
	Cross-validation	71	50	42	54	0.46
Metabolites						
Low	Re-substitution	100	50	0	13	0.59
	Cross-validation	100	50	0	13	0.59
Medium	Re-substitution	63	75	63	50	0.38
	Cross-validation	63	75	13	13	0.59
High	Re-substitution	13	50	63	88	0.47
	Cross-validation	38	50	63	88	0.41
All disease severity levels	Re-substitution	21	50	58	54	0.54
	Cross-validation	21	50	54	42	0.58

Models for low disease severity

Of the nine significant mass ions, selected by univariate analysis from a total of 150 ions (46–195 *m/z*), their abundance was subjected to a stepwise discriminant analysis which selected two ions upon which discriminant analysis (DA) models were developed. The DA models (re-substitution) correctly classified 13, 63, 25 and 88% of the observations into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively; the remaining observations were mis-classified with a total posterior probability error estimate rate of 53% (Table 2). Interestingly, the classification rate based on cross-validation of the DA models was exactly the same as that of the DA models based on re-substitution. The DA models classified the *L. theobromae* and W-control treatments better than the other treatments.

Models for medium disease severity

The abundance of 24 significant mass ions, selected by univariate analysis from a total of 150 ions, were subjected to the stepwise discriminant. The DA models correctly classified 57, 38, 63 and 25% (error rate 54%), and the cross-validation correctly classified 57, 38, 63 and 13% (error rate 59%) of the observations into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively (Table 2). The DA models for the N-control and *C. gloeosporioides* treatments were better than those for the other treatments.

Models for high disease severity

In the case of high disease severity, 74 significant mass ions, selected by univariate analysis from a total of 150

mass ions, were subjected to stepwise DA, which in turn selected 28 ions which were used to develop DA models. The DA models correctly classified 100% of the observations into their respective treatments, and cross-validation correctly classified 100, 100, 100 and 88% of the observations into the *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively, with an error rate of 0.03% (Table 2).

Models for all disease severity levels

The abundance of 52 significant mass ions, selected by univariate analysis from a total of 150 mass ions, were further reduced to seven based on stepwise DA, which in turn were used to develop DA models. The DA models correctly classified 71, 58, 50 and 71% (error rate = 38%) and the cross-validation correctly classified 71, 50, 42 and 54% (error rate = 46) of the observations into the *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively, and the remaining observations were mis-classified (Table 2). The DA model for the *C. gloeosporioides* treatment was better than the others for disease/inoculation discrimination. The DA models for high disease severity were better than those for the low, medium or all combined disease severities, in terms of the classification of unknowns (Table 2).

Test validation

Test validation of DA models for high disease severity correctly classified 25, 25, 25 and 88% of the fingerprints based on the low disease severity group for *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively, and 71, 38, 75 and 25%, respectively,

Table 3 Percentages of volatile metabolic fingerprints correctly classified by the best DA (re-substitution) models (based on high disease severity) as non-inoculated (N-control) or inoculated with water (W-control), *Colletotrichum gloeosporioides* or *Lasiodiplodia theobromae*, and when the DA models were test-validated using metabolic fingerprints not used in the model development

Models (disease severity)	DA-model/validation	<i>C. gloeosporioides</i>	<i>L. theobromae</i>	N-control	W-control	Error
Mass ions						
High using Low	Re-substitution	13	38	25	88	0.59
	Cross-validation	25	25	25	88	0.59
High using Medium	Re-substitution	50	38	50	13	0.63
	Cross-validation	71	38	75	25	0.48
Metabolites						
Medium using Low	Re-substitution	88	38	0	0	0.69
	Cross-validation	100	50	13	0	0.59
Medium using High	Re-substitution	25	25	13	63	0.69
	Cross-validation	0	75	25	75	0.56

of the observations based on the medium disease severity group. The DA models for high disease severity predicted medium disease severity better than low disease severity (error rates were 48% for low and 59% for the medium disease severity) (Table 3).

Models to discriminate diseases/inoculations based on abundance of metabolites

Metabolic fingerprints based on the normalized values of 31 metabolites (unique metabolites excluded) were grouped into (i) low, (ii) medium, (iii) high and (iv) all disease severity groups, based on disease severity.

Models for low disease severity

Three significant metabolites based on univariate analysis were reduced further to two (chloroform and ethyl acetate) based on stepwise DA. The DA models correctly classified 100, 50, 0 and 13% (error rate = 59%) of the observations into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively, and the classification rate was similar to that for cross-validation (Table 2).

Models for medium disease severity

Six significant metabolites, selected by univariate analysis from a total of 31 metabolites, were reduced further to three (2,6-dimethyl-2,4,6-octatriene, α -phellandrene and styrene) based on stepwise DA, and these were used to develop DA models. The DA models correctly classified 63, 75, 63 and 50% (error rate = 38%) and the cross-validation correctly classified 63, 75, 13 and 13% (error rate = 59%) of the observations into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively (Table 2).

Models for the high disease severity

For high disease severity situations nine significant metabolites, selected by univariate analysis from a total of 31 metabolites, were reduced further to two (ethanol and 2-methylfuran) based on stepwise DA. The DA models correctly classified 13, 50, 63 and 88% (error rate = 47%)

of the observations into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively, with a similar classification based on cross-validation (Table 2).

Models for all disease severities

Univariate analysis selected 7 significant metabolites from a total of 31, and these were reduced further to three (chloroform, ethyl propanoate and styrene) based on stepwise DA. The DA models correctly classified 21, 50, 58 and 54% (error rate = 54%) and cross-validation correctly classified 21, 50, 54 and 42% (error rate = 58%) of the observations into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively (Table 2). The DA models for medium disease severity were better than those for low, high or all disease severity levels in classifying unknowns (Table 2).

Test validation

Test validation of the best DA models, based on the medium disease severity group, correctly classified 100, 50, 13 and 0% of the observations based on low disease severity and 0, 75, 25 and 75% of the observations based on high disease severity into *C. gloeosporioides*, *L. theobromae*, N-control and W-control treatments, respectively (Table 3).

Discussion

In this study, a technology was developed to detect and discriminate four inoculations/diseases, including two diseases of mangoes, using a GC/MS platform, based on four different volatile metabolic profiling criteria. These were: (i) metabolites unique to an inoculation/disease; (ii) metabolites common to a few but not to all inoculations/diseases; (iii) metabolites common to all inoculations/diseases, but with relatively large differences in abundance among different inoculations/diseases; and (iv) discriminant analysis models based on metabolic fingerprints using normalized abundance of mass ions or relatively consistent metabolites.

This is the first study to provide data on the composition of the headspace volatile metabolites of mangoes inoculated

with pathogens and provides the basis for discriminating the postharvest diseases caused by *C. gloeosporioides* and *L. theobromae*.

Several compounds were unique to a disease/inoculation, which could be qualitatively used to discriminate diseases studied here, in unknown disease samples. 1-Pentanol and ethyl boronate were unique to *L. theobromae*-inoculated mangoes, which could be discriminated from *C. gloeosporioides* ones, which also produced a unique compound, thujol. 1-Pentanol and ethyl boronate were also reported to be unique for bacterial soft rot of carrot (Vikram *et al.*, 2006).

Disease-specific metabolites have been detected in other diseased fruits and vegetables. In a study on apples, methyl acetate was found to be unique to fruits inoculated with *Botrytis cinerea*, 4-methyl-1-hexene to fruits inoculated with *Mucor piriformis*, 2-methyltetrazole and butyl butanoate to fruits inoculated with *Penicillium expansum*, and 3,4-dimethyl-1-hexene and fluorethene to fruits inoculated with *Monilinia* sp. (Vikram *et al.*, 2004a). Acetyl hydrazide, propylcarbamate, propenyl bromide, acetone, 1-ethenyl-4-ethyl benzene, thiirane and 1-(methylthio)-E-1-propene were unique to onion bulbs inoculated with *Botrytis allii*, while 3-bromo furan was specific to bulbs inoculated with *E. carotovora* subsp. *carotovora* (Prithiviraj *et al.*, 2004). Also, 4-mercapto-3-(methylthio)-c-(thiolactone)-crotonic acid and 1-oxa-4,6-diazacyclooctane-5-thione were unique to *Fusarium oxysporum*-inoculated onions (Prithiviraj *et al.*, 2004). Seven unique compounds, viz. 1-pentanol, 3-methylbutanol, 2-methylpropanol, 2,3-butanedione, ethyl boronate, isopentyl methyl ether and ethane ethoxy were detected in carrots (cv. Vita Treat) inoculated with *E. carotovora* subsp. *carotovora* (Vikram *et al.*, 2006). The use of unique compounds for disease discrimination may be valid if the lesions are spatially separated, but its use when the diseases occur together in the same lesion remain to be validated.

Several metabolites were common to a few but not all treatments. The metabolites 1-butanol, styrene and ethyl propanoate were common to both *C. gloeosporioides*- and *L. theobromae*-inoculated mangoes, but were absent in both the controls. The absence of these compounds in healthy mangoes agrees with the findings of Narain & Galavao (2004). Methyl octanoate was detected only in *C. gloeosporioides*-inoculated, N-control and W-control fruits, but not in *L. theobromae*-inoculated ones. Thus, the presence and/or absence of the above compounds could be considered for qualitative discrimination of *L. theobromae* and *C. gloeosporioides*, especially when unique compounds are absent and mixed infections, especially in the same lesion, are present.

Some of the above compounds have been observed in other studies. Styrene was reported from potato tubers inoculated with *Fusarium coeruleum* and *Phytophthora infestans* (De Lacy Costello *et al.*, 2001). Methyl octanoate, which was detected here in mango fruits inoculated with *C. gloeosporioides*, and in N-controls and W-controls, was also reported from healthy fruits of the same cultivar (Narain & Galavao, 2004).

Several compounds were produced in significantly higher amounts in mangoes inoculated with pathogens than in controls. Even though these compounds were common to all the treatments, significant differences in their relative abundance can help to detect and discriminate diseases/inoculations, especially in the absence of unique or other disease-discriminatory compounds. Compounds such as ethanol, ethyl butanoate, ethyl 2-methylbutanoate, ethyl octanoate, ethyl 2-butenate and methyl (Z)-2-butenate varied significantly in their abundance and were therefore disease/inoculation-discriminatory.

The high abundance of ethanol in diseased mangoes when compared to controls can be used to discriminate diseased from non-diseased mangoes. Trace amounts ethanol production were reported in overripe mangoes stored for a long time (MacLeod & Snyder, 1985). Ethanol, a well known anaerobic metabolite, was also produced in high abundance in orange fruits inoculated with *Penicillium digitatum* (Pesis & Marinansky, 1990).

A total of 104 different aroma volatile compounds was detected in cv. Tommy Atkins (Narain & Galavao, 2004), including 14 terpenes (72.30%), 20 alcohols (6.36%), 23 esters (8.43%) and 11 other compounds (3.23%). 3-Carene and α -pinene were the most abundant compounds in the present study and were also reported as the most abundant by other workers (MacLeod & Snyder, 1985; Malundo *et al.*, 1997). Several other metabolites detected here, such as limonene, α - and β -caryophyllene, terpinolene, β -copaene, α -terpinene, 2-methylfuran, ethyl acetate, toluene and camphene, have also been reported by Narain & Galavao (2004).

The occurrence of a given volatile metabolite was inconsistent among different replicates. Some compounds were detected only in one out of eight replicates. The inconsistency of exogenous metabolites (headspace gas) among replicates was also reported in earlier studies on other crops (Prithiviraj *et al.*, 2004; Vikram *et al.*, 2004a,b; Lui *et al.*, 2005). Such variation is also not unusual in endogenous metabolic profiling studies (Roessner *et al.*, 2001; Dixon *et al.*, 2002). Some inconsistencies are associated with the misidentification of metabolites using the NIST library, especially using mass ions in the limited range of 46–300 *m/z*. In the present study, the headspace gas was accumulated for a few hours, while dynamic metabolic profiling may have partially avoided such inconsistencies. Reactions among different volatiles and also between volatiles and fruits or vegetables have been reported as other potential reasons for variability in volatile profiles among replicates (Hamilton-Kemp *et al.*, 1996). However, 34 compounds were still relatively consistent among replicates and these were used to develop criteria for detecting and discriminating diseases of mangoes in storage.

Since disease discrimination based on metabolites is not always possible because of inconsistencies in their occurrence, a modelling approach was considered. Some volatile metabolites were not detected in the early stages of disease development. Accordingly, models were developed separately for different disease severity levels. In general,

models based on the fingerprints of the abundance of mass ions were better than those based on the abundance of the relatively consistent compounds, possibly because of the error associated with the identification of compounds and the non inclusion of compounds with relatively low abundance ($<10^4$) when deriving fingerprints based on metabolites. Furthermore, unique compounds (the best qualitative discriminators) were not included when deriving fingerprints based on the abundance of metabolites. In spite of these limitations, cross-validation analysis correctly classified 100 and 50% of the observations into *C. gloeosporioides* and *L. theobromae* treatments, respectively, for low disease severity, and 63 and 75% for medium disease severity. It should be mentioned that in addition to classification based on discriminant analysis models, the presence of unique compounds that were excluded from the model, as they were not present any other treatments, must be considered in the final correct classification.

Discriminant analysis models correctly classified up to 100% of the observations, depending on the treatment. Multivariate analysis has also been used to discriminate different abiotic and biotic factors responsible for milk spoilage (Marsili, 1999) and metabolic phenotypes (Roessner *et al.*, 2001). It is preferred that the models developed should be validated using a dataset not used in model development. For the present study, the validation based on a jackknife procedure (cross-validation) was unbiased. It also properly tested the predictive ability of the model in question (Johnson, 1998). In addition, the DA models were test-validated, using a dataset not used in developing the model.

Of the four metabolic profiling criteria developed here, that based on unique compounds is the best. However, since the occurrence of these unique compounds is inconsistent, other criteria also should be considered. Discriminatory compounds, qualitative or quantitative, will increase the probability of correct classification. In addition to these, models also can be used to further increase that probability. Models based on the relative abundance of mass ions are very easy to develop and can be automated for user-friendly applications, whereas models based on compounds require correct identification of compounds.

The technology developed in this study could be used by store managers to make intelligent management decisions, such as how long the product can be stored, or which lots can be kept for longer periods. For commercial applications, the samples could be enclosed in airtight containers for 2.5 h at room temperature to accumulate volatiles and GC/MS (Hapsite model or other) can be used to trap and analyse volatiles, although this is time-consuming and laborious as samples need to be collected. However, to collect headspace gas directly from the stores more large-scale studies are required, and background volatiles, depending on the storage sanitation, may make this less practical. A portable GC/MS equipped with a pre-concentrator was used in the present study in the hope that technology transfer would be easy, as sampling, gas analysis and disease detection could be done onsite.

Alternatively, headspace gas can be collected using tube traps, and the metabolites desorbed either thermally or using organic solvents, and analysed using any other suitable GC/MS (Lyew *et al.*, 1999). Pre-concentration of headspace volatiles with tube traps would be necessary to detect the low levels of volatiles present in commercial storage headspace. When extending this technology to industrial scales, careful attention should be paid to the effect of cultivars (Vikram *et al.*, 2004b). If there is significant cultivar effect, then separate models for each cultivar or group of cultivars with similar discriminatory compounds are required. In the present study, the experimental units were contained in a bottle and the volatiles were sampled separately for each disease to avoid cross-contamination. Although outbreaks of one disease in storage are quite common, they may often be associated with low levels of other diseases. Criteria based on inoculation/disease-discriminatory metabolites can be used to detect when a mixture of diseases is present, although their validity must be tested before recommendation.

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